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Total Number of Pages: 02

Course: B.Tech, IDD (B.Tech and M.Tech)
Sub_Code: REC3C002

3rd Semester Regular/Back Examination: 2023-24

SUBJECT: Signals and Systems

BRANCH(S): ECE, ETC

Time: 3 Hour

Max Marks: 100

Q.Code: N508

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part-I

Q1 Answer the following questions: (2 x 10)

- How to check stability of a discrete time system?
- Determine the DFT of $\delta(n) + \delta(n-1)$.
- $y(n) = x(n) + x(n-1)$ is time-invariant system or not.
- Differentiate between convolution and correlation of two discrete time signals.
- Write any two properties of discrete time convolution.
- Differentiate between linear and circular convolution.
- Write any two properties of z-transform.
- Write the formula for a continuous time Fourier Series expansion.
- What is circular convolution and how it is different from linear convolution?
- Define the superposition theorem to check linearity of a system.

Part-II

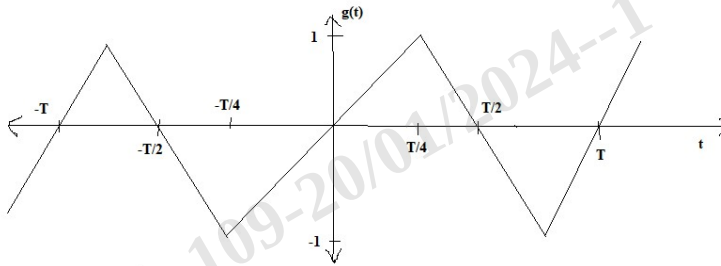
Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 x 8)

- Determine the Z-transform of $x(n) = \{-3, 1, 2, 0, 0, 0, 1, 5, 9\}$
- Find the circular convolution of $x(n) = \{1, -1, 3\}$ & $h(n) = \{1, 0, -1\}$
- Check the linearity and time-invariance of the systems represented by

$$h[n] = y[n-1] + \frac{2y[n]}{y[n-3]}$$

- Determine the DFT of $2\delta(n) + 3\delta(n-1) + 3\delta(n-4)$.
- Find the z-transform of $x(n) = \cos \omega_0 n$ for $n \geq 0$.
- Using the properties of Z transform, find the Z transform of $3^n u[n-3]$.

- g) Find the fundamental term of the Fourier series of the waveform shown below.



- h) Find the inverse Z-transform of $H(z) = \frac{1 + 2z^{-1}}{1 - 4z^{-1} - 2z^{-2}}$.
- i) Find the DTFT of the sequence $x(n) = u(n - k)$.
- j) Derive the relationship between Fourier transform of $x(2t + 3)$ and that of $x(t)$.
- k) Find the linear convolution of $x(n) = \{2, 4, 6, 8\}$ and $h(n) = \{1, 2, 3, 4\}$ using the properties of Z-Transform.
- l) Why do we need both Fourier Series and Fourier transform to analyze the signals in frequency domain? Justify your answer with suitable examples.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

- Q3** Compute the DFT of $x[n] = \cos \frac{n\pi}{4} + \sin \frac{n\pi}{4}$ where $0 \leq n < 8$. (16)
- Q4** Find the inverse Z-Transform of $H(z) = \frac{z}{3z^2 - 4z + 1}$ using partial fraction expansion method for all possible ROCs. (16)
- Q5** Compute the continuous time Fourier transform of the signal $x(t) = \begin{cases} 1, & \text{for } -4 \leq t \leq 4 \\ 0, & \text{for } 4 \leq |t| \leq 10 \end{cases}$ (16)
- Q6** Write Short Notes (Any Two)
Frequency Domain Sampling
Inverse Z-Transform
Properties of DFT (8x2)